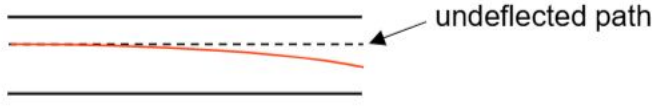
4(a)(i)	The magnetic force is downwards (by FLHR, B points into the paper and current is to the left) in the region between the plates, hence the electric force must be upwards. Since the particles are negatively charged, the electric field must be downwards (opposite to the electric force). Hence, P is positive. (Plate Q is grounded, hence is at 0 V.)	**B1** **B1**
4(a)(ii)	To move undeflected, the net force must be zero. $F_E = F_B$ $qE = qvB$ $v = \frac{E}{B}$	 **M1** **A0**
4(a)(iii)	 $v > \frac{E}{B} \Rightarrow qvB > qE$, or $F_B > F_E$. Since the resultant force points downwards, the particles will be deflected downwards. (Note that the path is not parabolic.)	**B1** **B1**
4(b)(i)	The magnetic force is always perpendicular to the velocity. Hence, it does no work on the particles. The kinetic energy, hence the speed, of the particle remain constant.	**B1** **B1**
4(b)(ii)	The magnetic force provides the centripetal force. $qvB = m\frac{v^2}{r} \Rightarrow \frac{q}{m} = \frac{v}{Br}$ With $v = \frac{E}{B}$, (show substitution) $\frac{q}{m} = \frac{E}{B^2 r}$	**M1** **M1** **A0**